Welcome

Studying transitions with the ONS Longitudinal Study

Workshop

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This presentation is from our experience as data users. It draws on many helpful materials created by the Celsius user support team which are cited throughout - any errors remain our own.

Housekeeping: fire exits · toilets · Wi-Fi = eduroam · refreshments · **break around 11:25**

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Who we are

We're from the **Centre for Employment Research** at the University of Westminster, a research centre working on education and labour market economics, economic evaluation (RCTs), and employee relations.

Emma Gorman
Principal Research Fellow

Min Zhang
Research Fellow

Peter Urwin
Professor of Applied Economics and Director of CER

- **ADR UK Community Catalyst – Youth Transitions, and an ADRUK affiliate community**
- The aim is to build capacity, and a community, around using administrative data, particularly for studying transitions
- This is ongoing: running events like this to share **experience as data users** – 'nuts and bolts'

What we want to do today -

10:00	Welcome, who we are, and introductions	All
10:20	The ONS LS: content, structure and access	Emma
10:50	Constructing data and variables	Min
11:25	Break	
11:40	Case studies: longitudinal analysis and panel data models	Emma
12:15	Discussion (if time): your research questions	Emma, Min, Peter
13:00	Close – and happy to have further discussions afterwards	

Times are approximate.

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All

Introductions

ACTIVITY | about 10 minutes

Tell us:

- Your name and where you work
- Anything specific you are wanting to gain from this session?

The ONS Longitudinal Study: an introduction to the data

ONS LS workshop 2026

Emma Gorman
Centre for Employment Research, University of Westminster

What I'll cover

1. **What the LS is**
   Who is in it, why it exists, who uses it

2. **How the data are structured**
   Linked records, tables, households, cross-sections and panels

3. **Linkage, and what has changed**
   Who gets followed up

4. **How it compares**
   Against other UK data, and its strengths and weaknesses

5. **Accessing the LS**
   Ways in, how it has changed, and how long things take

A 1% sample of England and Wales Census, since 1971

~1%
of the population of England and Wales

4
undisclosed birthdays define membership

6
censuses linked: 1971 to 2021

1.3m
traced members since 1971

Dynamic 1% sample of Census records, linked longitudinally at individual level - plus **life events** linked at the individual level.

- Managed by the fantastic **Celsius user support team** at UCL, and maintained by the LS team at the ONS

Sources: Shelton et al. (2018); ONS (2026) Completeness of census linkage.

Why it was set up in 1974

Problem 1: occupational mortality

Deaths came from death certificates and the population from the census, so the two didn't match. The last job on a death certificate was a poor guide to lifetime occupation.

Problem 2: fertility

The early-1970s 'birth dearth' prompted demand for better data on birth spacing and family influences on fertility.

Solution: link census records to births, deaths and cancer registrations for the same people, via the NHS Central Register

Source: Shelton et al. (2018), 'Why was the cohort set up?'

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Who uses it

Epidemiology and public health
Mortality and cancer inequalities

Demography
Fertility, migration, household change

Sociology
Social class, mobility, ethnicity

Economics
Labour markets, education, intergenerational outcomes

Geography
Internal migration, small areas

Government
Health inequalities, pensions, care

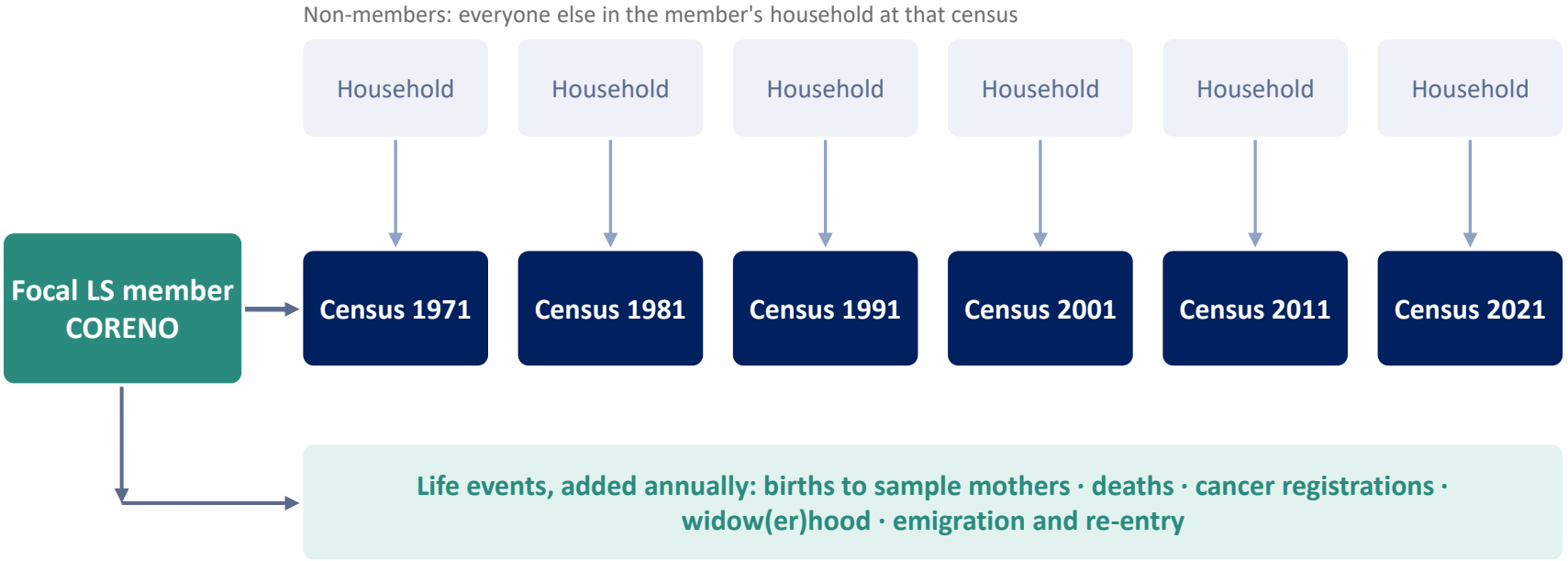
It has fed into major policy reviews



Navy: health inequalities. **Teal: pensions and care.**

Source: Shelton et al. (2018), 'What has it found?'

Focal LS member is followed across Censuses



Tables are joined on CORENO: member census (ME), non-member (NM) and events tables

Sources: Shelton et al. (2018); CeLSIUS Defining a Study Population guide (2025); CeLSIUS data dictionary.

ONS LS structure: six censuses, linked life events

↓ Entry events, 1971–2021
New births 368,417 · Immigrations 296,668 · Re-entries 26,755

↑ Exit events, 1971–2021
Deaths 321,492 · Embarkations 58,003 · Enlistments 7,608

1971	1981	1991	2001	2011	2021
N = 529,900	N = 536,141	N = 543,885	N = 540,063	N = 585,161*	N = 648,383*

In every census

Sex
Age
Household composition
Marital / legal partnership status
Household amenities
Economic activity
Industry / occupation
Hours worked
Residential location
Country of birth

Added later
new questions, by census

+ 1981
Work location

+ 1991
Ethnic group
Long-term limiting illness

+ 2001
Self-rated health
Care-giving
Religion (voluntary)

+ 2011
Passports
Year of arrival in UK
National identity
Main language

+ 2021
Sexual orientation (voluntary)
Veteran status

Other linked life events

Live births to LS mothers (345,826)
Widow(er)hoods (104,047)
Still births to LS mothers (1,934)
Cancer registrations, to 2023 (160,727)
Infant deaths to LS mothers (2,418)

** LS members present with full data. Source: adapted from Dr Aly Sizer, CeLSIUS, UCL (ONS LS presentation, YouTube).*

What the files look like

Member census tables

One row per member, one table per census (1971 … 2021)

CORENO	Age	Tenure	Class
1001	6	Owner	–
1002	34	Social	IV
1003	71	Owner	II

Non-member tables

One row per other person in the member's household

CORENO	Relation	Age	Class
1001	Mother	33	III
1001	Father	35	II
1002	Partner	36	IV

Events tables

One row per event, added from registration data

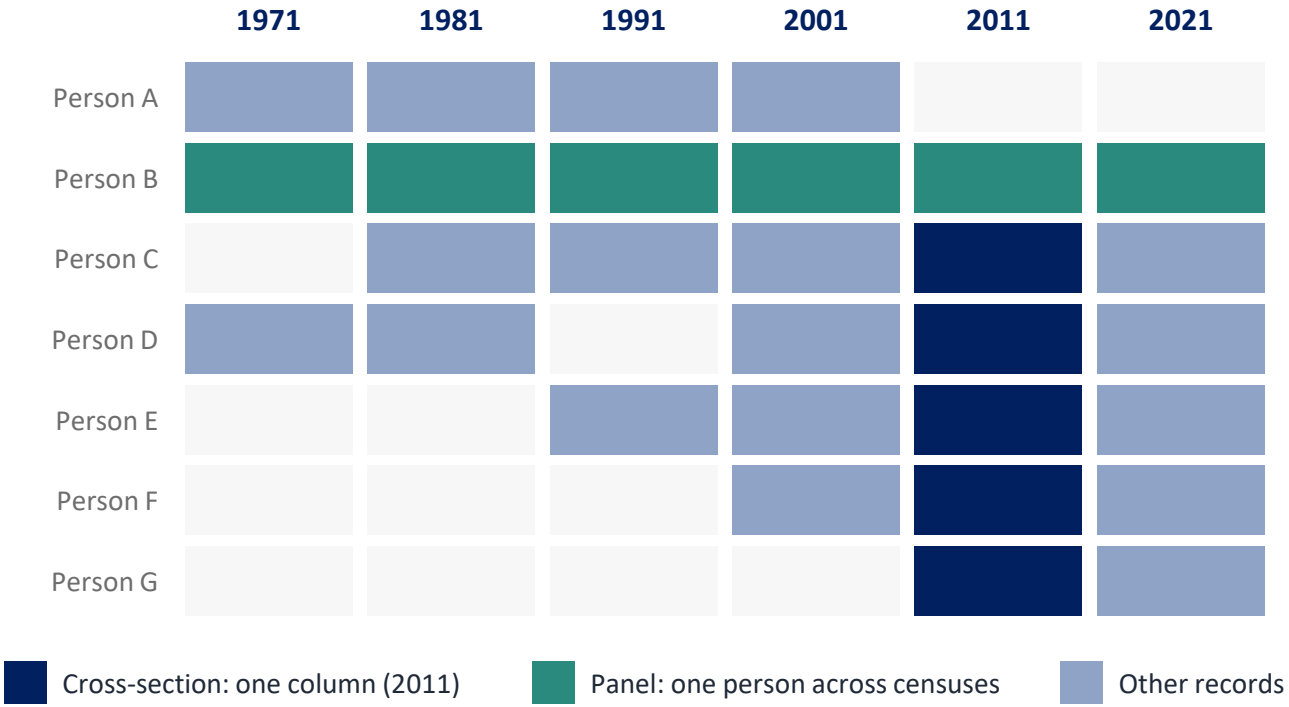
CORENO	Event	Year
1003	Death	2014
1002	Birth	2016
1002	Cancer	2019

Everything joins on **CORENO**, the member's LS number.

Member 1001 is a child in this census. You see their parents through the non-member table, and the same CORENO picks them up again at later censuses.

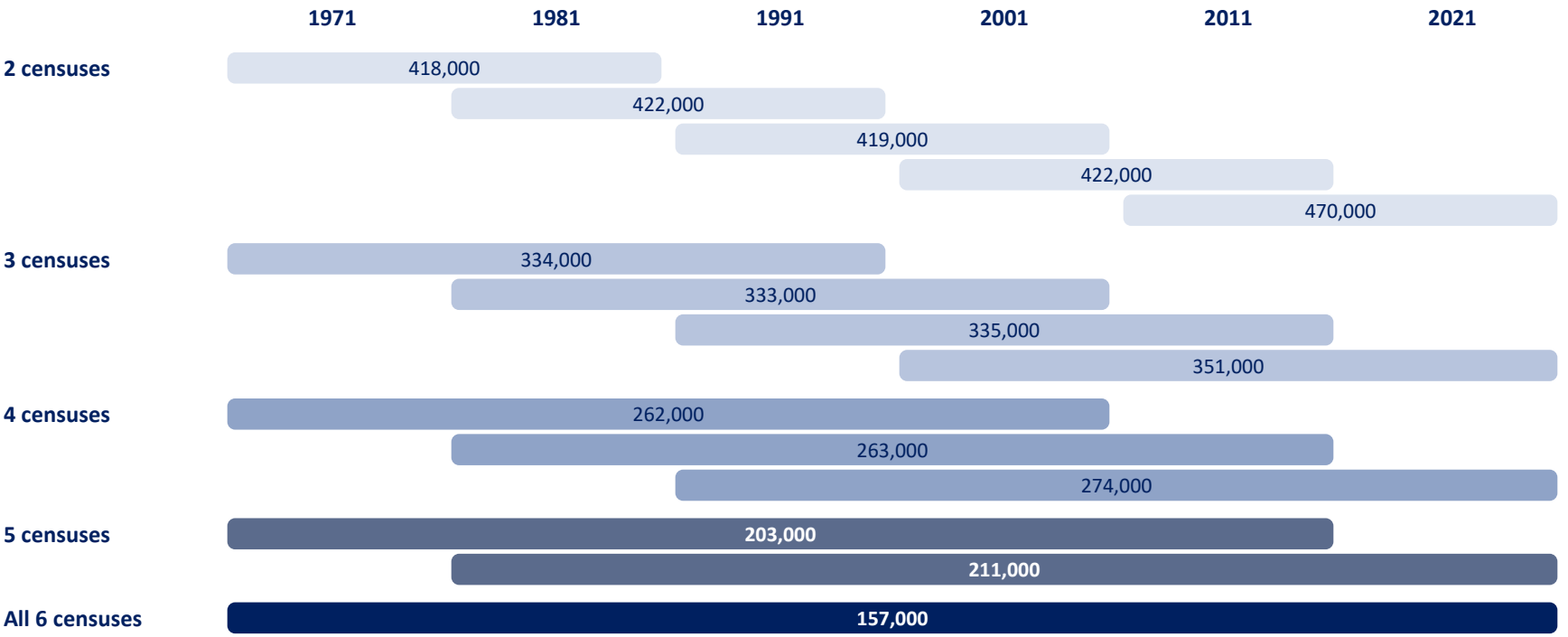
Illustrative: made-up values and simplified variable names. Real tables are wider. See the CeLSIUS data dictionary and LIDS.

Constructing your sample: cross-sections or as a panel?



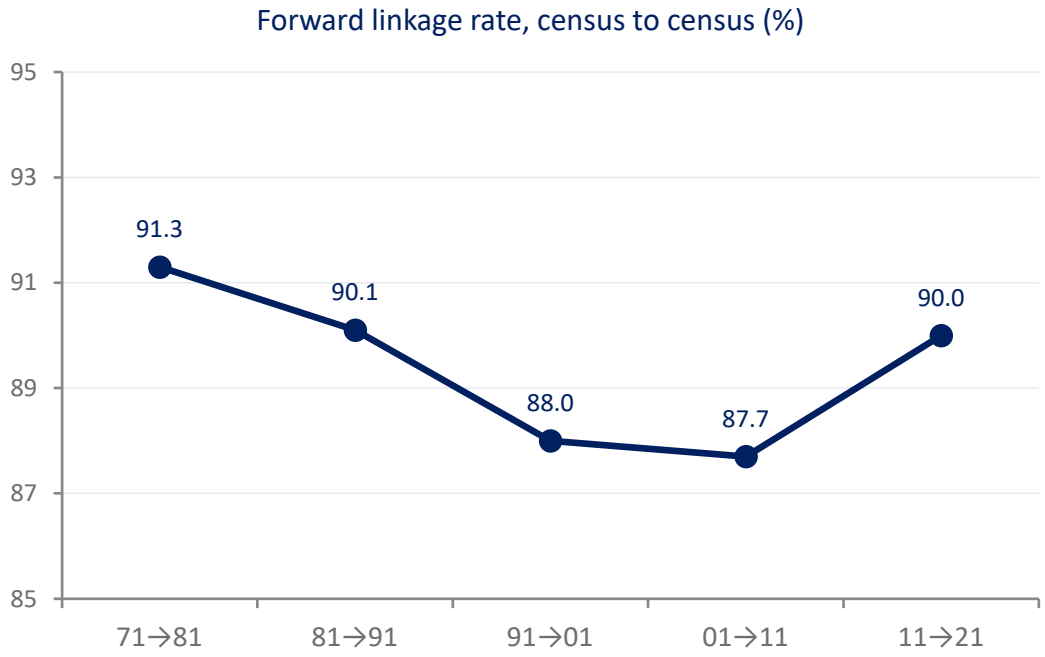
Illustrative. Based on the three study types in the CeLSIUS Defining a Study Population guide (2025).

How many people can you follow across censuses?



Sample sizes rounded to the nearest 1,000. Source: adapted from Dr Aly Sizer, CeLSIUS, UCL (ONS LS presentation, YouTube).

About nine in ten are found again at the next census



High linkage rates

Source: ONS (2026) Completeness of census linkage, Tables 4 to 6.

Codes and classifications change over time



Occupation coding at each census

Social class

Registrar General's Social Class until 2001, then NS-SEC

Qualifications

1971–1991: only those gained after age 18. From 2001: GCSE/O-level up

Cause of death

ICD-8 (1971–81), ICD-9 (1981–2000), ICD-10 (2000 onwards)

Sources: CeLSIUS Socio-Economic Indicators guide (2025); Shelton et al. (2018). SOC2020 for 2021: ONS.

How it compares with other UK data

	ONS LS	Understanding Society	Birth cohorts	Labour Force Survey
Sample	~1% of E&W; >1m ever	~40,000 households at start	~17–19k per cohort	Large, rotating
Frequency	10-yearly + annual events	Annual	Irregular sweeps	Quarterly, 5 waves
Span	1971–2021	2009– (BHPS 1991–)	1958, 1970, 2000–01 births	Short panel
Income	**No**	Yes	Yes	Earnings, waves 1 and 5
Attrition	Low (~90% linked)	Survey attrition	Survey attrition	By design
Coverage	Everyone, incl. communal	Private households	Varies	Private households

Simplified. Sources: Shelton et al. (2018); study websites

Strengths and weaknesses

Strengths

Size	Small groups, areas, occupations
Length	Fifty years, six censuses
Coverage	Whole population, incl. **communal establishments (e.g., care homes)**
Low attrition	~90% linkage
Households	e.g., parents, partners, siblings

Weaknesses

Infrequent	Ten-yearly snapshots
Thin content	No income or behaviours
Changing measures	Questions and codes shift
'Narrow' dataset	But can match in extras
Access	**Secure setting;** cells ≥ 10

Sources: Shelton et al. (2018), 'Main strengths and weaknesses'; ONS (2026).

Ways to access the LS via ONS Secure Research Service

1 Safe settings

Go to a secure room

- ONS safe rooms in Titchfield and Newport
- SafePods, mostly at universities, and SafePoint at LSE
- Open to any accredited researcher with an approved project; book (far) ahead

2 AOC: office access

Use the SRS from your own organisation

- Your organisation signs an Assured Organisational Connectivity (AOC) agreement
- Access from an approved secure office, no booking
- The organisation has to meet ONS security standards

3 AOC: home access

Use the SRS from home

- Your organisation needs approval for remote access
- Organisation's VPN and approved device only
- Only from the home address you registered with ONS

Source: ONS, Accessing the Secure Research Service (ons.gov.uk).

How our own access to the LS has changed



The old days

- Send remote jobs to the CeLSIUS team, who ran them for us
- Or travel to the ONS data rooms in Newport or London

COVID

- Data rooms closed
- Back to sending remote jobs to the CeLSIUS team

AOC agreement

- Westminster's assured organisational connectivity agreement
- Access from our own approved office

Now - home access

- Agreed subject to conditions

Our experience at Westminster. Arrangements differ between organisations.

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Clearance and timelines: what to expect (and Menti)

Step	What it is	Who	Typical time	Our tips
Data application	After accredited researcher status - a project application WITH CeLSIUS, to be approved by ONS	You, CeLSIUS, ONS	Variable! 3+ mo.	Ensure you request everything you need from the start
Intermediate clearance	Checks on working outputs so you can take them out of the SRS. Only for people on your project	CeLSIUS	Very quick with Celsius checking	
Publication clearance	Final check before results can be shared outside the project team: e.g., papers, slides, reports	CeLSIUS / LS team at the ONS	Variable – 1 day to 2 weeks+	
Code clearance	Getting your code out of the SRS, e.g. to share or reuse	Celsius/ONS	Variable	
Look-up table ingest	Bringing in **lookup** files, e.g. area lookups (NOT data)	Celsius	Usually quick, 1-2 days	Need to be clear about the purpose of lookup and ensure all permissions have been gained

The LS is different: outputs go through intermediate clearance with CeLSIUS and then publication clearance with ONS-LS team and Celsius. Most other SRS datasets have one step: ONS clears them for publication.

Getting started

First steps

- Become an ONS accredited researcher
- Talk to CeLSIUS (user support team!) early about feasibility and variables Apply for your project together with CeLSIUS
- Technical access - Find out whether your organisation has AOC or if you will need to use saferoom/safepods

Prepare in advance

- **Longitudinal Impossible Dataset (LIDS)**: fake data with the real structure, to write and test code
- CeLSIUS data dictionary and thematic guides
- Shelton et al. (2018) cohort profile

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Helpful links

CeLSIUS (UCL): the LS support team

celsius@ucl.ac.uk
User support: feasibility, variables, applications, clearance

CeLSIUS website
About the LS, data dictionary, guides, how to apply

Longitudinal Impossible Dataset (LIDS)
Synthetic data with the real LS structure, to test code

LIDS explained (IJPDS article)
Background on how LIDS was built and how to use it

ONS Secure Research Service

Accessing the Secure Research Service
Safe rooms, SafePods, AOC office access and home access

Become an accredited researcher
Eligibility, Safe Researcher Training and how to apply

Apply for an accredited research project
The ONS project application process

srs.connectivity@ons.gov.uk
Check whether your organisation has AOC, or start an application

Also useful: SafePod Network · SRS support: srs.customer.support@ons.gov.uk · CALLS Hub (UK census longitudinal studies) · Shelton et al. cohort profile (IJE)

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Over to you

Questions?

About the data, or about getting access

Next: Min Zhang on constructing data and variables for longitudinal analysis

Challenges and options of data harmonisation

ONS LS workshop 2026

Min Zhang
Centre for Employment Research, University of Westminster

Why harmonise variables over time?

Parents, children, cohorts

Origins and destinations are measured decades apart, often on different classifications.

Cohorts are compared across fifty years.

Change within people

Fixed-effects and panel models use change between censuses.

We need consistent coding across different

Linking area data

Outside data, such as earnings by local authority, must sit on the same areas as the LS.

The aim: a variable that means the same thing at every census you use.

Examples in this talk draw on our ESRC-funded work on intergenerational social mobility using the ONS LS.

Challenges of harmonisation

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		Example in this talk
1	**Who is asked** Routing changes who answers a question	**Occupation of the unemployed**
2	**Coding frame** The classification is replaced	**CO70 to SOC2020**
3	**Categories** Answer options split, merge or are renumbered	**Housing tenure**
4	**Boundaries** Areas are redrawn	**Local authority districts**

Identifying the unemployed: the categories move

Census	Variable	Unemployed categories
1971	ECONP7, STUJOB7	Out of employment, sick; out of employment, other. Seeking work (*STUJOB7*)
1981	ECONACT8	Waiting to take up a job; seeking work. Temporary sickness is a separate box
1991	ECONPO9	Waiting to start a job; unemployed
2001	ECOP0	Seeking work and available within 2 weeks; waiting to start (ILO-based, ages 16–74)
2011	ECOP11	Seeking and waiting combined into one code
2021	ECOP21	Seeking and waiting combined into one code

Lowest common denominator: seeking and waiting can only be told apart up to 2001, so combine them.

Source: CeLSIUS data dictionary, member tables ME71–ME21.

Who is asked about their occupation?

1971	1981	1991	2001	2011	2021
Most recent job if retired or out of work	Most recent full-time job	Most recent job	Last main job	Last main job	Last main job
No time limit	No time limit	Only if within last 10 years	Any time; year last worked	Any time; year last worked	Any time; only a 12-month flag

Teal = narrower than the other censuses

Housewives are routed out

1971 and 1981 exclude housewives without a job, even with past work. This affects mothers in origin households.

Check it in the SRS

Cross-tabulate the unemployed against a valid occupation code at each census before choosing a rule.

Sources: Census forms 1971–2021, England household forms (UK Data Service; ONS); ONS (2021) occupation question development.

Every census uses a different occupational classification

1971	1981	1991	2001	2011	2021
CO70	CO80	SOC90	SOC2000	SOC2010	SOC2020
221 groups	348 groups	371 groups	353 groups	369 groups	412 groups
OCCS7	OPOCCST8	SOC909	OCCP0	OCCP11	OCCP21

Groups: unit groups (CO80: operational codes), excluding special codes. Variable: LS member table.

Not one-to-one

Groups split and merge between versions, so recoding needs a crosswalk, not a relabel.

Source: CeLSIUS data dictionary; author counts of codes listed for each variable.

Occupations change, not only their codes

Occupation	1971 CO70	1981 CO80	1991 SOC90	2001 SOC2000	2011 SOC2010	2021 SOC2020
Bus conductors	✓	✓	✓	Leisure and travel service n.e.c.	Leisure and travel service n.e.c.	Leisure and travel service n.e.c.
Telegraph and radio operators	✓	✓	✓	Communication operators	Communication operators	Communication operators
Compositors	✓	✓	✓	✓	Pre-press technicians	Pre-press technicians
Coal mining	✓	✓	✓	✓	✓	Mining and quarry workers
Computer programmers / software	—	✓	✓	✓	✓	✓
Call centre agents	—	—	—	✓	✓	✓
Web design and development	—	—	—	—	✓	Web design professionals
Cyber security professionals	—	—	—	—	—	✓
Data analysts	—	—	—	—	—	✓

✓ own group. — no separate group. **Shaded:** new occupations.

ONS reasons for revision: SOC2010 tightened the definition of managers; SOC2020 reviewed professional vs associate professional roles and reclassified IT.

Sources: CeLSIUS data dictionary code lists (OCCS7 to OCCP21); ONS relationship tables; ONS User Guide 2010:22, App. A; ONS SOC 2020 Vol. 1, s. 1 and 4.

Reading the ONS relationship tables

Per old code

Share of its cases in each new code; the shares across new codes sum to 100

By sex

Separate columns for men and women

By sample

Samples differ: SOC2000 2411 → 2413, men: 67%, 73%, 81%

Thresholds

Older guides drop links <5% and flag codes with <5 cases (*)

ONS: the relationships are "provided as a guide only" to those you will find in your own data.

Three ONS tables: SOC90 → SOC2000 and SOC2000 → SOC2010 (User Guides 2000:22, 2010:22); SOC2010 → SOC2020 (July 2021 workbooks).

Sources: ONS User Guides 2000:22 and 2010:22 (sections 2–4, Appendix B); ONS SOC2010–SOC2020 relationship tables (July 2021).

Worked example: SOC2000 to SOC2010

One code, split three ways: SOC2000 2411 → SOC2010

SOC2000 2411
Solicitors, lawyers, judges, coroners

	Men	Women
2412 Barristers and judges	17.0%	11.7%
2413 Solicitors	**73.3%**	**79.3%**
2419 Legal professionals n.e.c.	9.7%	9.0%

2001 Census 1% sample. Each column sums to 100. Most common shaded.

Most common

Everyone → 2413 Solicitors. Right for 73% of men, 79% of women.

Use the shares

Apply fractional weights. Keeps the split.

Move up a level

Minor group 241, legal professionals, is an exact match.

Source: ONS User Guide 2010:22, Table 1a (Unit), 2001 Census 1% economically active sample.

How we did it: the most common destination, by sex

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1 Look up the ONS table
a many-to-many relationship table

2 Take the mode, by sex
the new code with the highest share, for men and for women

3 Assign it
to every LS member of that sex with the old code

Example: SOC90 872 Drivers of road goods vehicles → SOC2000

SOC2000 destination	Men	Women
8211 Heavy goods vehicle drivers	**58.7%**	8.9%
8212 Van drivers	38.4%	**85.1%**

Mode, shaded: men → 8211 HGV drivers, women → 8212 van drivers. 1991 Census sample: 2,228 men, 101 women.

Why by sex

The same old code can have a different share for men and women.

Compromises

The mode matches the ONS coding only for its share: here 59% of men, 85% of women.

Source: ONS User Guide 2000:22 (SOC90 to SOC2000), Table 1a (Unit), 1991 Census sample.

Housing tenure: the answer options keep changing

	1971	1981	1991	2001–2021
Owner occupied	1 owner occupied	1 freehold 2 leasehold	1 mortgage 2 outright	0 outright 1 mortgage 2 shared ownership
Social rented	2 council or New Town	3 local authority 5 housing association	4 council 5 New Town / HAT 6 housing association	3 council 4 other (social rented)
Private rented	3 private unfurnished or housing association 4 furnished	6 furnished 7 unfurnished	7 furnished 8 unfurnished	5 landlord or agency 6 employer 7 relative or friend 8 other (private rented)
Other	0 not stated	4 with job or business	3 with job or business	9 rent free

1971: housing associations coded as private renting.

Code 0: "not stated" in 1971, "owns outright" from 2001.

Source: CeLSIUS data dictionary (TENURE7, TENURE8, TENURE9, TENH0, TENH11, TENH21). Labels shortened.

Relationship to LS member: same code, new meaning

Code	1971	1981	1991	2001	2011	2021
3	Father	Child-in-law	Son or daughter	Child	Partner	Partner
4	Mother	Parent	Child of cohabitee	Stepchild	Child	Son or daughter
8	Unrelated / outside family	Grandchild	Parent-in-law	Grandchild	Parent	Mother or father

Variables: LSRELAT7, NLSRELT8, NLSRELT9, LSRELAT0, LSRELAT11, LSRELAT21 (non-member tables)

Note: 1991 has 43 codes, of which 24–43 are ambiguous combinations (e.g. 26 'Parent, other unrelated', 39 'Parent, uncle or aunt'); they cover only a small number of observations.

Sources: CeLSIUS data dictionary (NM71–NM21); CeLSIUS 1991 Census Appendix 32 (LSRELAT9).

Local authority boundaries: within the LS and for linkage

403 districts, 1991 → **376** districts, 2001 → **348** districts, 2011

England and Wales, LS member tables

Within the LS

The LS supplies district codes harmonised to 1991 boundaries, plus 2001, 2003 and 2011 versions.

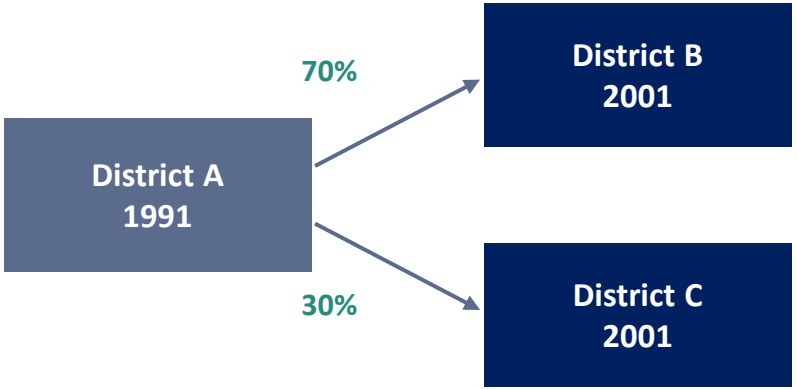
Linking outside data

ASHE 2001 uses a different district definition from LS 2001. Put both on one geography before merging.

Two approaches: population-weighted conversion, or matching by name.

Source: CeLSIUS data dictionary (LAD91CD, LAD91CDC, LAD91CDO, LAD01CD, LAD03CD, LAD11CD).

Approach 1: population-weighted conversion



rate(A) = 0.7 × rate(B) + 0.3 × rate(C)

e.g. 0.7 × 6.0% + 0.3 × 4.0% = 5.4%

Weights are shares of A's population; they sum to 1.

1. **Population shares**
   of each source district in each target district (GeoConvert)
2. **95% rule**
   if ≥95% falls in one target, treat as unchanged
3. **Weighted sum**
   otherwise, Σ share × value
4. **Merge**
   onto the LS districts

Assumes the rate is uniform within B and C, and uses 2011 populations for older boundaries.

Illustrative numbers. Weights: GeoConvert (UK Data Service), 2011 Census postcode populations.

Approach 2: matching by name

LS LAD 91 name	Updated name	Manipulation
Westminster	City of Westminster	Cleaning
Stoke-on-Trent	Stoke on Trent	Cleaning
St. Helens	St Helens	Cleaning
Woodspring	North Somerset	Rename
Brighton	Brighton and Hove	Merging
Hove	Brighton and Hove	Merging
Lliw Valley (60%)	Swansea	Merging with compromises
Swansea (100%)	Swansea	

Cleaning

Punctuation, spacing, "City of": same area

Rename

New name, same area

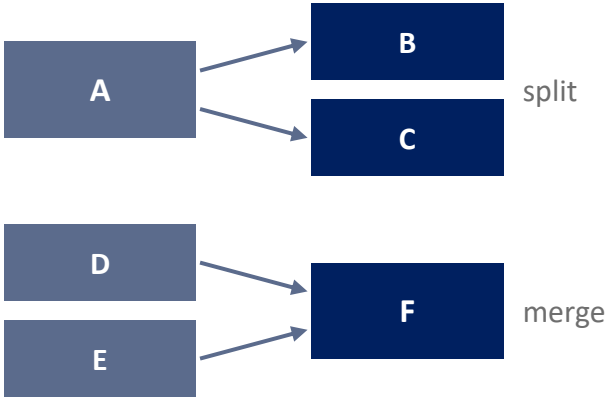
Merging

Two or more 1991 districts → one district later

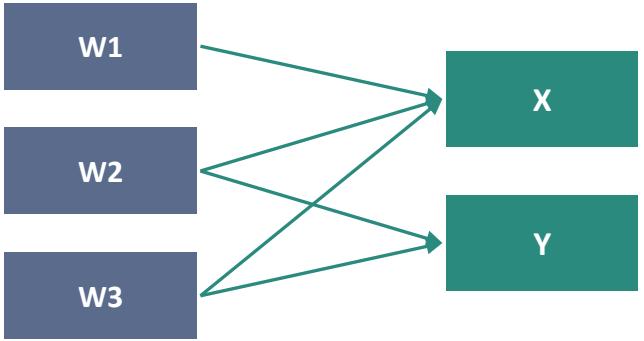
Source: authors' matching of LS 1991 district names.

Wales is harder than England

England: usually a split or a merge



Wales 1996: many-to-many



Illustrative

Source: author test on district labels in the CeLSIUS data dictionary (LAD91CDC, LAD01CD).

A recipe for harmonising

1. **Define the concept**
   what you need to mean the same thing at every census
2. **Read the routing**
   who was asked, not just which codes exist
3. **Find the lowest common denominator**
   e.g. owner vs renter; seeking and waiting combined
4. **Use bridges and document**
   dual-coded years, crosswalks; keep the raw variables
5. **Test sensitivity**
   re-run key results under the alternative choices

AT YOUR TABLES

Share your experiences:

Which variable

do you need, and at which censuses?

Where does it break?

Routing, coding frame, categories or boundaries

What is the common denominator

that holds at every census you use?

Will you link outside data,

and on which geography?

Next: break, then Emma's case studies of panel analyses with the LS

Social mobility and policy evaluation: ONS LS case studies

ONS LS workshop 2026

Emma Gorman
Centre for Employment Research, University of Westminster

Roadmap: three parts

PART 1 · MEASUREMENT

Building mobility measures, step by step

- Birth cohorts and follow-up
- Parents from the household records
- NS-SEC, dominance and comparisons

PART 2 · EXEMPLARS

What the measures can show

- Place: geographic mobility, BJS (2021)
- Time: Children of the 1990s (2026)
- Wealth: housing mobility, JSP (2025)

PART 3 · POLICY EVALUATION

Grammar schools and social mobility

- The policy issue and the question
- What we matched in
- How the LS brings it together; findings

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PART 1

Building social mobility measures, step by step

Emma Gorman

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Step 1: define birth cohorts, then follow them at every census

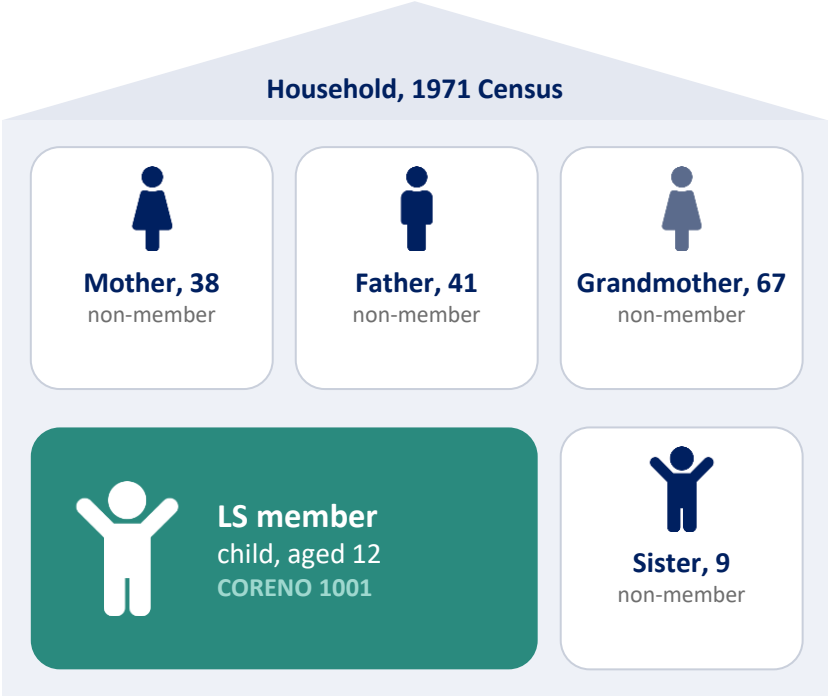
Born	1971	1981	1991	2001	2011	2021	Follow-up
1954–63 aged 8–17 in 1971	Age 8–17	Age 18–27	Age 28–37	Age 38–47	Age 48–57	Age 58–67	50 years
1964–73 aged 8–17 in 1981		Age 8–17	Age 18–27	Age 28–37	Age 38–47	Age 48–57	40 years
1974–83 aged 8–17 in 1991			Age 8–17	Age 18–27	Age 28–37	Age 38–47	30 years
1984–93 aged 8–17 in 2001				Age 8–17	Age 18–27	Age 28–37	20 years
1994–2003 aged 8–17 in 2011					Age 8–17	Age 18–27	10 years

Origin census: aged 8–17, usually still living with parents

Follow-up: the same person's own census record, linked by CORENO

Cohort definitions follow Buscha, Gorman, Sturgis & Zhang (2026). The 1994–2003 cohort shows what the 2011–2021 link makes possible. Ages are approximate, at census day.

Step 2: the census records everyone in the household



Member file (ME81) *one row per LS member*

CORENO	Age	Tenure	Class
1001	12	Owner	–

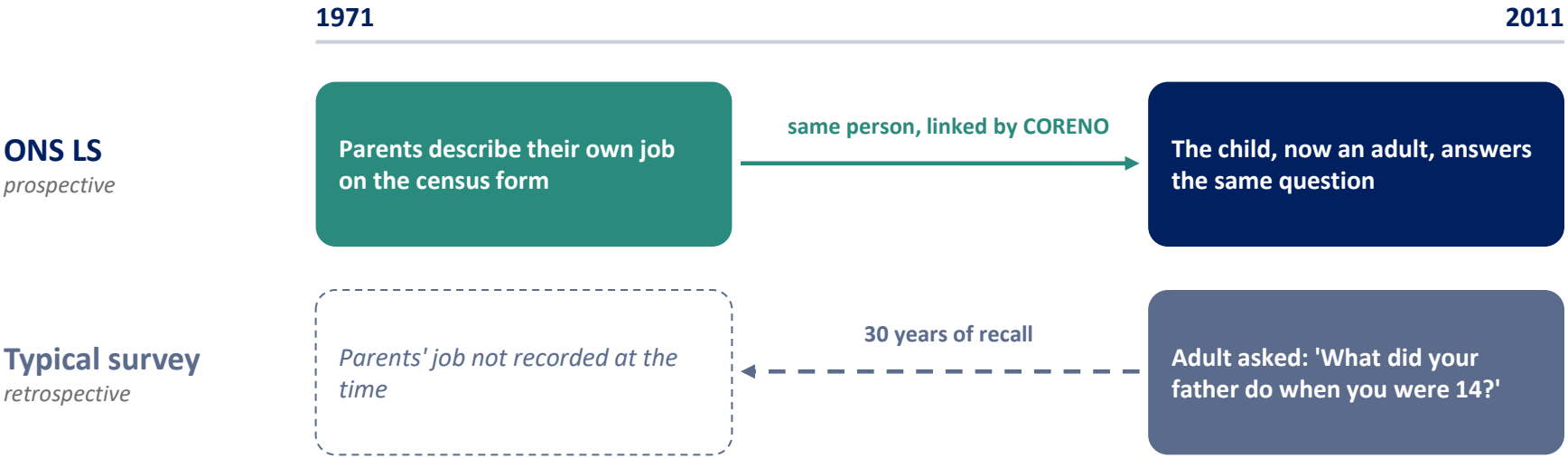
Non-member file (NM81) *one row per other person*

CORENO	Relationship	Age	Occupation
1001	**Mother**	38	Nurse
1001	**Father**	41	Electrician
1001	Sister	9	–
1001	Grandmother	67	Retired

Same CORENO = same household. The relationship code picks out parents, and siblings or grandparents if we want them.

Illustrative: made-up values and simplified variable names. Real tables are wider; see the CeLSIUS data dictionary.

Step 3: parents' jobs are observed at the time, not recalled



No recall error, no 'upgrading' of parents

Mum's and dad's jobs both observed directly

Similar question and coding for parent and child

Retrospective origin questions are used in, for example, Understanding Society and the Labour Force Survey. See also Buscha, Gorman & Sturgis (2021, 2023).

Step 4: code jobs to NS-SEC and take the higher-class parent

1	Higher managerial & professional
2	Lower managerial & professional
3	Intermediate
4	Small employers & own account
5	Lower supervisory & technical
6	Semi-routine
7	Routine

Mum: nurse

Dad: electrician

THE DOMINANCE APPROACH

Origin = the higher of the two parents' classes

Mum's NS-SEC 2 outranks dad's NS-SEC 5, so this child's origin is NS-SEC 2.

OTHER CHOICES

Father's class only (the traditional approach), mother's class only, or each parent separately.

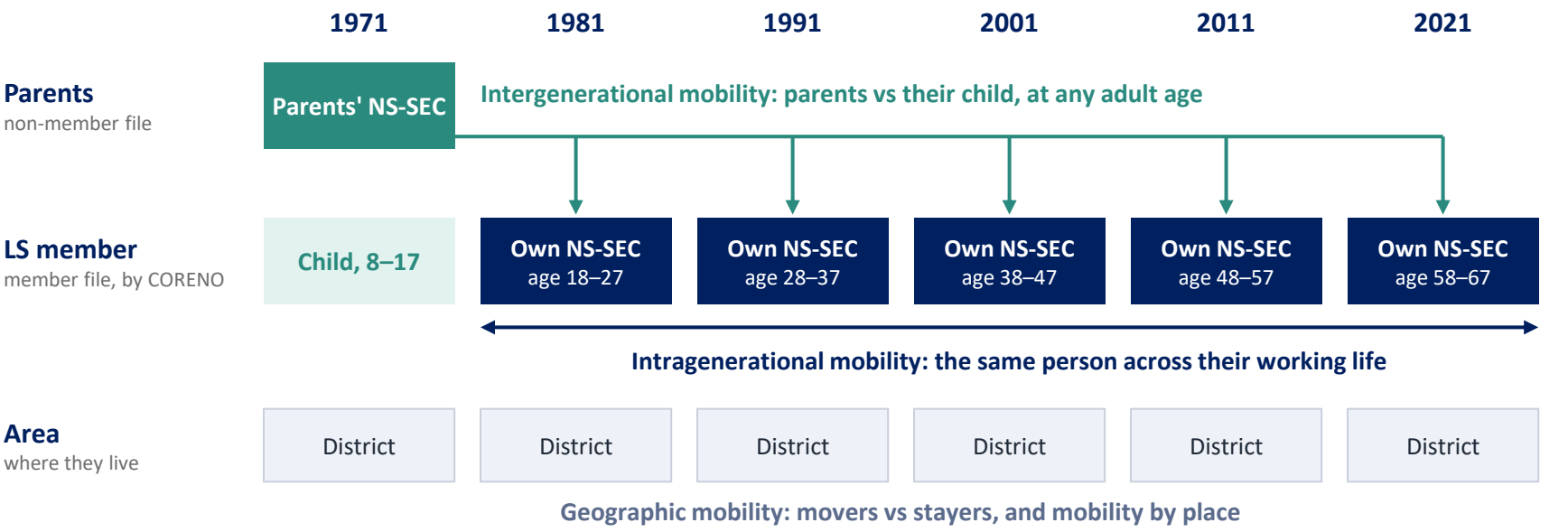
FOR HEADLINE MEASURES

Two groups: NS-SEC 1–2 vs 3–7 (professional & managerial vs the rest)

NS-SEC analytic classes 1–7; class 8 (never worked, long-term unemployed) excluded. Dominance approach: Erikson (1984). Occupations are illustrative.

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Step 5: zoom out and compare origins with every follow-up



Repeat for every cohort and compare at the same age (e.g. 28–37): trends in mobility across generations

Cohort shown: born 1954–63 (aged 8–17 in 1971), followed to 2021. Same census questions and coding for parents and children.

Measuring mobility: absolute and relative

ABSOLUTE MOBILITY

What share of people end up in a higher (or lower) class than their parents?

- A simple percentage: e.g. the share moving from working-class origins into professional or managerial jobs

RELATIVE MOBILITY

How much do your origins shape your chances?

- Compares the chances of reaching the top for people from advantaged and disadvantaged origins (an odds ratio)

In our papers: 'high' = NS-SEC 1–2 (managerial and professional), 'low' = NS-SEC 3–7; origin = the higher-class parent; destination = own class at 28–37.

Following Erikson & Goldthorpe (1992); Bukodi & Goldthorpe (2018). Robustness in the papers: 7-class NS-SEC, CAMSIS scores, linear probability models.

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Recap: what the LS solves, and what it doesn't

	Retrospective survey	ONS LS
Origin measure	Adult recalls parents' job 'when you were 14', decades later	Parents' own census returns, recorded at the time
Measurement error	Recall and social-desirability error; not classical, so bias of unclear sign	No recall; same classification for parents and children
Missing origins	Absent or unknown parents often unrecorded	Both co-resident parents observed; dominance rule
Precision	A few thousand per cohort: no areas or minority groups	≈50,000 per cohort; ~400 districts; ethnic groups
Attrition	Cumulative across waves	≈10% per decade, not cumulative

THE COSTS

Origins only for children living with a parent at the origin census · no income or earnings · NS-SEC for 1971 and 1981 rebuilt via SOC90 without employer size · destinations at 28–37 risk life-cycle bias · a 1% sample means small cells and output checks

Sources: Buscha, Gorman & Sturgis (2021, 2023); Buscha et al. (2025, 2026); Shelton et al. (2019).

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PART 2

Exemplars: what the measures can show

Emma Gorman

Examples

EXEMPLAR 1

Geographic mobility

Does mobility differ between places? Do people who move do better?

COMPARES

Parents' class → own class, by region and district; movers vs stayers

BJS (2021)

EXEMPLAR 2

Time trends

Has mobility changed for people born in the 1990s?

COMPARES

Parents' class → own class at 28–37, across four cohorts

Children of the 1990s (2026)

EXEMPLAR 3

Housing mobility

Do ethnic groups differ in passing on home ownership?

COMPARES

Parents' tenure → own tenure, by ethnic group

JSP (2025)

ALSO POSSIBLE

Educational mobility

Do children reach higher qualifications than their parents?

COMPARES

Parents' → own qualifications; best from 2001

Work by others

PART 3

Policy evaluation

Did grammar schools change social mobility?

COMPARES

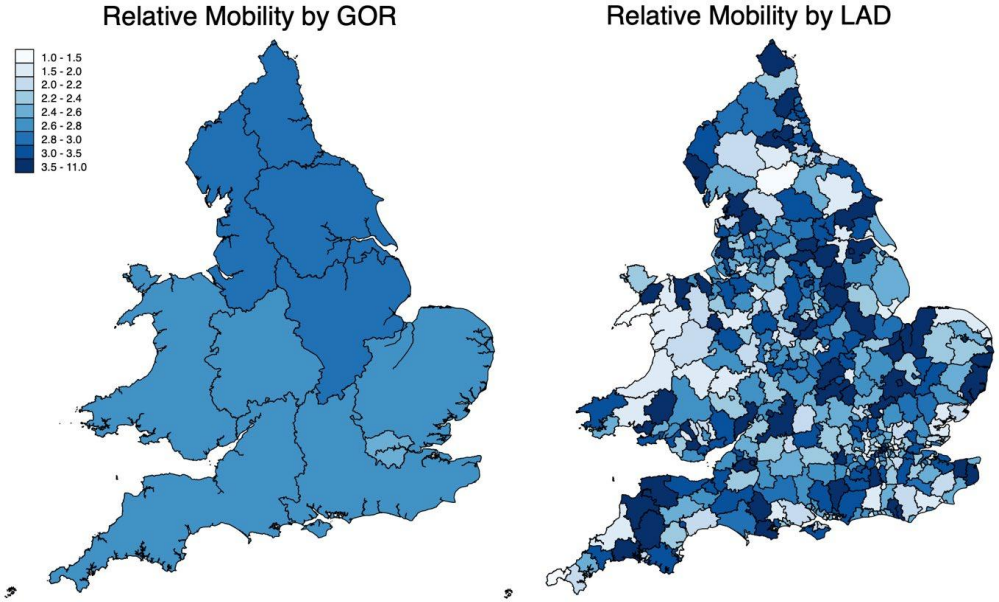
Mobility in areas with more vs less selective schools

Labour Economics (2023)

Same files, different cuts: the variable (class, tenure, education), the grouping (place, cohort, ethnicity) and the measure are all choices made after linkage.

Buscha, Gorman & Sturgis (2021, 2023); Buscha, Gorman, Sturgis & Zhang (2025, 2026). Full references at the end.

Place: big differences *within* every region



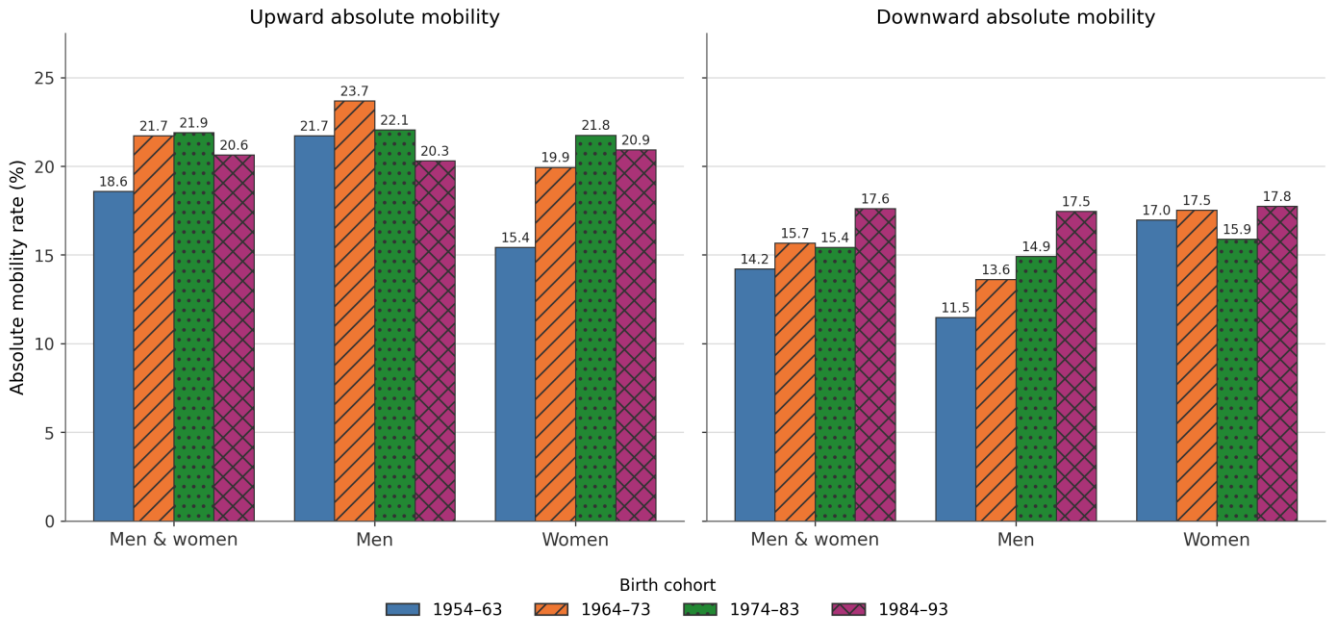
Odds ratios by region (left) and district (right), pooled over cohorts born 1953–83. Darker = less mobile.

FOR POLICY

A regional or North vs South lens misses low-mobility districts in affluent regions

Source: Buscha, Gorman & Sturgis (2021), British Journal of Sociology 72(5), Figure 3, Tables 2 and 4. N = 168,777.

Cohort trends



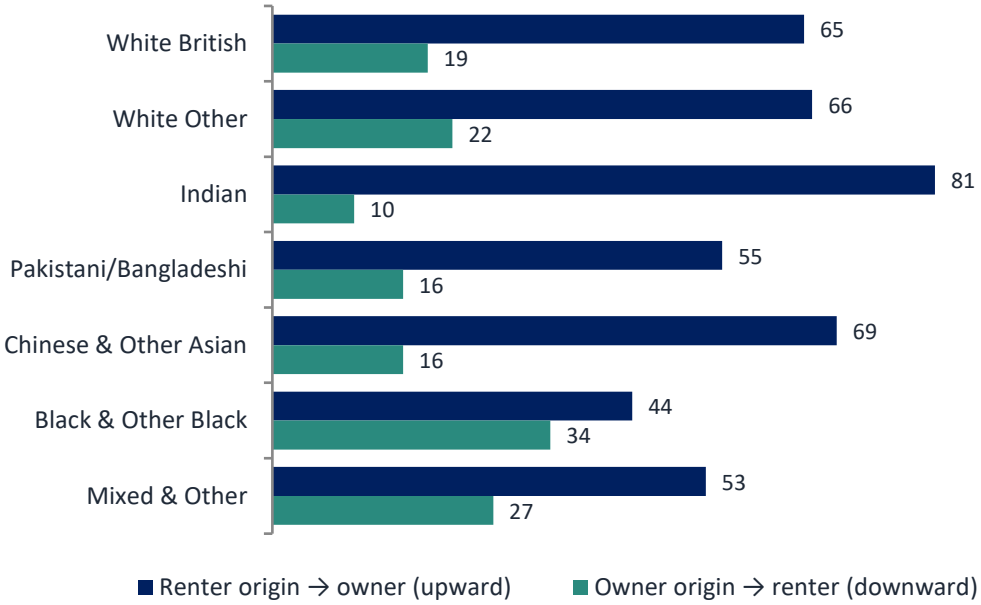
3.14 → 2.42

odds ratio, 1954–63 to 1984–93: more fluid

Source: Buscha, Gorman, Sturgis & Zhang, Children of the 1990s (unpublished seminar slides, Sept 2026). Binary NS-SEC, self-employed excluded.

Wealth: home ownership passes on unevenly

Share making each transition (%), cohorts born 1954–83 pooled



For those with renter parents, chance of owning?

For those with owner parents, chance of renting?

Source: Buscha, Gorman, Sturgis & Zhang (2025), Journal of Social Policy 54, Table 2. Origin at age 8–17; destination at 28–37.

3

PART 3

Policy evaluation: grammar schools and social mobility

- Using the LS to evaluate long-run policy changes

Emma Gorman

Does selective schooling causally increase social mobility?

1944	Education Act: 'tripartite' system, with the 11+ test allocating pupils to grammar, secondary modern or technical schools
1965	Circular 10/65 asks, but doesn't require, local education authorities (LEAs) to go comprehensive
1967–83	Selective schooling falls from near-universal to a minority, at different speeds in different LEAs
2017	Manifesto pledge to lift the ban on new grammar schools, to boost social mobility

Our question

What are the causal effects of the decline in the selective schooling system on intergenerational social class mobility in England?

Why the *system*, not the school?

Earlier studies compare pupils who just passed the 11+ with those who just missed. Selection also changes schools and peers for everyone else.

Source: Buscha, Gorman & Sturgis (2023), Labour Economics 81, 102336, sections 1 and 3.

Decline in English grammar schools

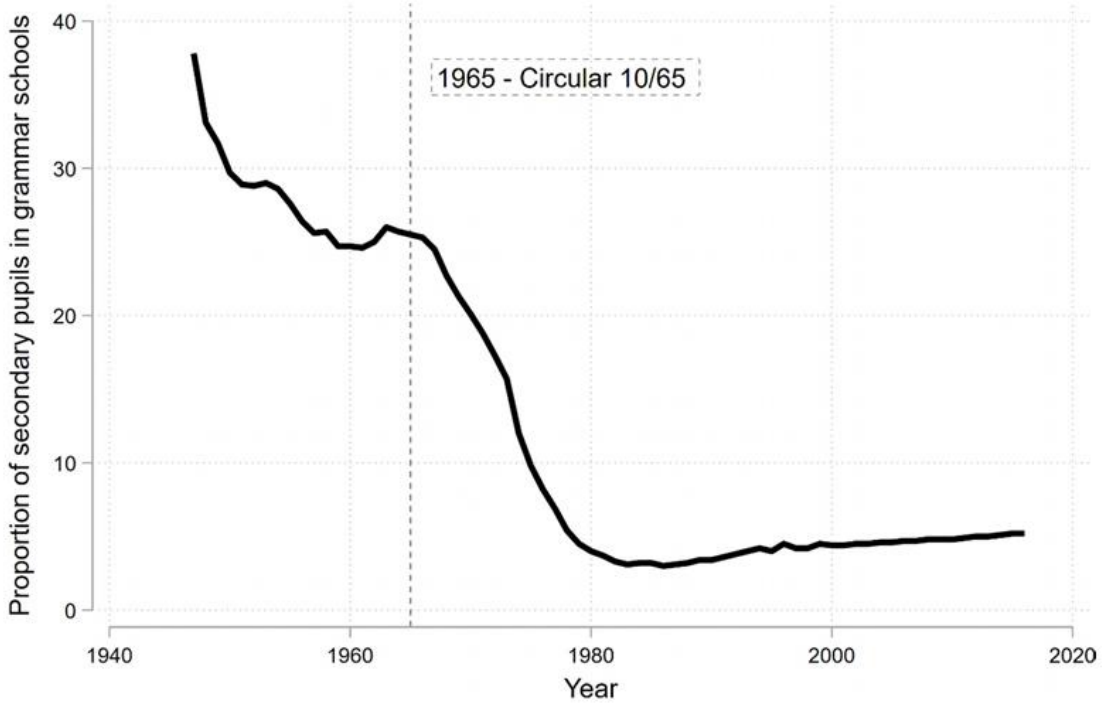
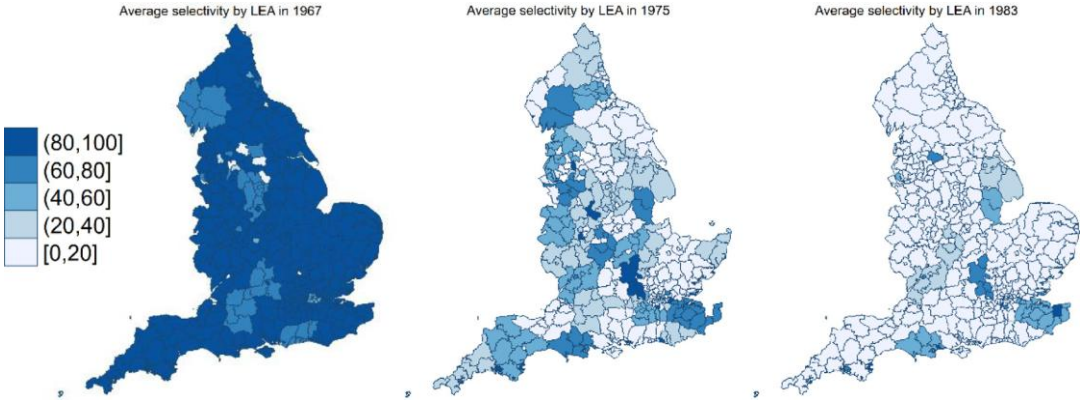


Fig. 1. Proportion of secondary school pupils taught in state-funded grammar schools, 1947 - 2016
Notes: Data source: Bolton, P. (2020). This Figure plots the proportion of secondary school pupils taught in state-funded grammar schools in England, 1947 – 2016.

Matched in the % in selective schools by area/time



Share of 13-year-olds in selective-system schools, by LEA: 1967, 1975, 1983. Darker = more selective.

Treatment $S_{gt} \in [0,1]$ is continuous, declines at different speeds, and starts from different baselines. Selective areas also differed at baseline.

141 LEAs × 17 years

Annual Schools Census, 1967–83, compiled by Prof Damon Clark (UC Irvine)

Same end point, different speeds

Most LEAs went from near 100% to near 0%

Why timing varied

Local logistics and population growth more than politics or mobility trends

Source: Buscha, Gorman & Sturgis (2023), Figure 3 and Table 2. 'Selective system' = grammar, secondary modern and technical schools.

The intuition: a staggered difference-in-differences

1

Areas moved away from selection at different times

Between 1967 and 1983 each LEA went comprehensive on its own timetable: some early, some late, a few hardly at all.

2

Compare changes, not levels

Cross-sectional and longitudinal variation allows us to adjust for area and time effects

Loose idea is that we can adjust for both area and time effects, and assume remaining variation is plausibly random
- much of the grammar school literature uses only cross-sectional variation (in the treatment)

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Identifying assumptions, intuitively

Assumption	In words	Threat in the grammar schools setting
Parallel trends	Without the policy, treated and comparison areas' mobility would have moved together	Timing of reorganisation correlated with local trends: population growth, political control, industrial decline
No anticipation	Outcomes don't respond before treatment starts	Families move towards or away from selective LEAs ahead of reorganisation; schools change intake early
No spillovers	One area's treatment doesn't affect another's outcomes	Cross-boundary schooling and 'cream-skimming' from neighbours (limited before the 1988 Act)
Stable composition	Repeated cross-sections compare like with like across cohorts	Selective migration; private schooling (≈6–7%); area-varying linkage failure
Homogeneous effects (for TWFE)	Same effect across areas and cohorts, or else use robust estimators	Effects may build up as a system matures, and differ by dose

Adapted from Roth, Sant'Anna, Bilinski & Poe (2023); Buscha, Gorman & Sturgis (2023), sections 3 and 6.

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The (old) workhorse: two-way fixed effects

$$Y_{gt(i)} = \alpha + \underbrace{\beta S_{gt}}_{\text{effect of selectivity}} + \underbrace{X'_{gti}\lambda}_{\text{controls}} + \underbrace{\gamma_g}_{\text{LEA FE}} + \underbrace{\delta_t}_{\text{cohort FE}} + \underbrace{\theta_g T_t}_{\text{LEA trends}} + \varepsilon_{gti}$$

Y	Social mobility of person i, growing up in LEA g, cohort t
S	share of pupils in selective-system schools: 0 = fully comprehensive, 1 = fully selective
γ, δ	LEA fixed effects absorb stable area differences; cohort trends to account for national trends
θT	Accounts for LEA-specific time trends

Use Jeffrey Wooldridge's *extended two-way fixed effects* framework for refinements to this model (new updates since 2023)

Source: Buscha, Gorman & Sturgis (2023), equations 1–2c. X: sex, parental age and age squared.

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Two-way fixed effects

Controls	Absolute upward mobility (mean 0.19)	Relative mobility, odds ratio (mean 3.08)
LEA fixed effects	**−0.050*** (0.005)**	**+0.59** (0.18)**
+ cohort fixed effects	−0.001 (0.007)	−0.43 (0.30)
+ LEA-specific trends	−0.009 (0.010)	−0.48 (0.42)

Coefficient on selectivity (0 = fully comprehensive, 1 = fully selective). LEA-clustered SEs. N = 90,894. A higher odds ratio = less mobility.

Conclusion: no evidence that either system changed aggregate mobility.

Area effects don't explain it

With LEA effects alone, selection looks harmful: about −5 points of upward mobility

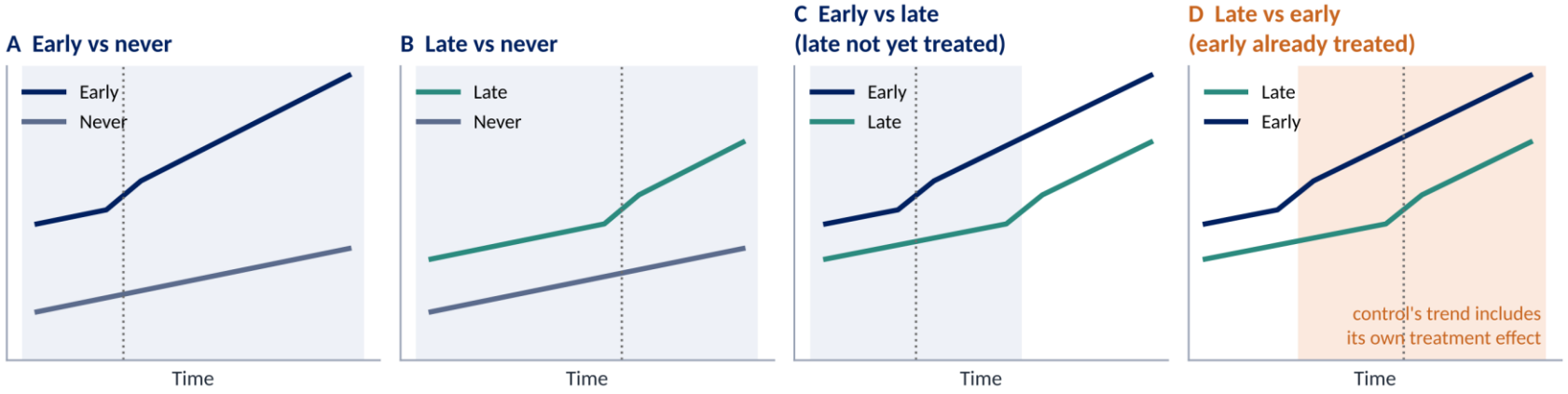
Common trends do

Mobility rose nationally as selection fell. Once cohort effects are in, the effect is close to zero

*Source: Buscha, Gorman & Sturgis (2023), Table 3. *** p < 0.001, ** p < 0.01.*

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Why TWFE can mislead with staggered adoption



A weighted average of 2 × 2s

$\beta_{TWFE} = \Sigma_k w_k \cdot DiD_k$ across all timing pairs (Goodman-Bacon 2021)

Weights aren't chosen by you

They depend on group sizes and on when in the panel treatment happens, not on policy relevance.

'Forbidden' comparisons

D uses already-treated units as controls. If effects grow over time, some ATTs get negative weight (de Chaisemartin & D'Haultfœuille 2020).

Stylised illustration: three adoption groups, effects that grow after adoption. Shaded = comparison window.

New approach: the de Chaisemartin–D'Haultfœuille estimator

HOW IT WORKS, INTUITIVELY

Only make 'clean' comparisons

- **Switchers**: LEAs whose selectivity changes between one cohort band and the next.
- **Stayers**: LEAs whose selectivity doesn't change over the same two cohort bands.
- Compare the change in mobility for switchers with the change for stayers, for each pair of cohorts, then average across periods.
- Already-treated areas never act as the benchmark for areas just changing, so there are no negative weights, and effects can differ across areas and over time.
- Inference by block bootstrap, clustered by LEA (100 replications)

Reference: "Causal inference – the remix" book from Scott Cunningham

https://mixtape.scunning.com/

Same conclusion

	TWFE	dCDH
Absolute upward mobility	−0.001 (0.007)	−0.004 (0.044)
Relative mobility (OR)	−0.432 (0.300)	−0.620 (1.431)

TWFE with LEA and cohort fixed effects. dCDH estimates slightly larger, both far from significant.

TODAY IT WOULD BE THE HEADLINE

Heterogeneity-robust DiD is now the main approach. The newer *did_multiplegt_dyn* (2024) adds dynamic, event-study effects for non-binary treatments that change gradually, as selectivity did.

de Chaisemartin & D'Haultfœuille (2020, 2024); Buscha, Gorman & Sturgis (2023), section 6.4 and Table B4.

Grammar schools: what we found

Raw association

Children who grew up in more selective areas had slightly lower social mobility.

Explained by time trends

The association disappears once we account for national and LEA-specific time trends

No effect either way

Neither selective nor comprehensive systems changed absolute or relative mobility. **Confidence intervals can rule out the extreme claims made by both 'sides' of the debate**

Robust

To other estimators, ways of measuring exposure, and mobility measures (details in backup).

INDIVIDUALS VS THE SYSTEM

Winners and losers

Grammar schools may well have helped some children from poorer backgrounds who got in.

But gains for those who passed the 11+ appear to be offset by losses for those sent to secondary moderns.

- **As a *system*, academic selection did not raise social mobility.**

Buscha, Gorman & Sturgis (2023), Labour Economics 81, 102336, sections 6–7.

The ONS LS for longitudinal analysis: strengths and trade-offs

THE LS'S STRENGTH

Captures **long-run social change** reliably: whole cohorts, every area, outcomes twenty years on, low attrition.

- Large enough to compare areas, cohorts and groups.
- Match in policy and area data on the right geography, and you can evaluate effects of important policies on long-run outcomes.

HARMONISATION OVER TIME

Questions, classifications and boundaries change between censuses.

A NARROW DATASET

Census questions only, every ten years: few variables, so mechanisms are hard to open up, and no year-by-year detail.

LEARNING FROM THE PAST

Long-run outcomes take decades to observe, so it says less about recent policy changes.

Shelton et al. (2019); ONS (2026), completeness of census linkage.

Thank you

THANK YOU

Thank you for attending!

Further questions?

Questions after today?

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University of Westminster

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COMMUNITY CATALYST: YOUTH TRANSITIONS

ADR England, on the ADR UK Learning Hub

adruk.org › Learning hub › User groups and communities

YOUTH TRANSITIONS RESOURCES

Guides, code for post-16 transitions in LEO/NPD

github.com/Youth-Transitions/resources/wiki

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